

CCNA(L1) PROJECT

Airport Network Design Using VLANs

1. Objective

To design and implement an airport network using VLANs, trunk links, and inter-VLAN routing for efficient departmental communication.

2. Software and Hardware Used

- Cisco Packet Tracer
- 1 Router
- 1 Core Switch
- 6 Access Switches
- PCs
- Servers (HTTP/FTP)
- UTP Cables

3. VLAN Configuration

VLAN	Department
10	Check-In
20	Security
30	Boarding
40	Baggage
50	Admin
60	Servers

4. Trunk Configuration

Trunk links connect the router, core switch, and access switches, allowing multiple VLANs to travel over a single connection.

5. Communication Between VLANs

VLAN	Department	Can Communicate With
10	Check-In	VLAN 50, VLAN 60
20	Security	VLAN 50, VLAN 60
30	Boarding	VLAN 60
40	Baggage	VLAN 50, VLAN 60
50	Admin	VLAN 10, VLAN 20, VLAN 40, VLAN 60
60	Servers	VLAN 10, VLAN 20, VLAN 30, VLAN 40, VLAN 50

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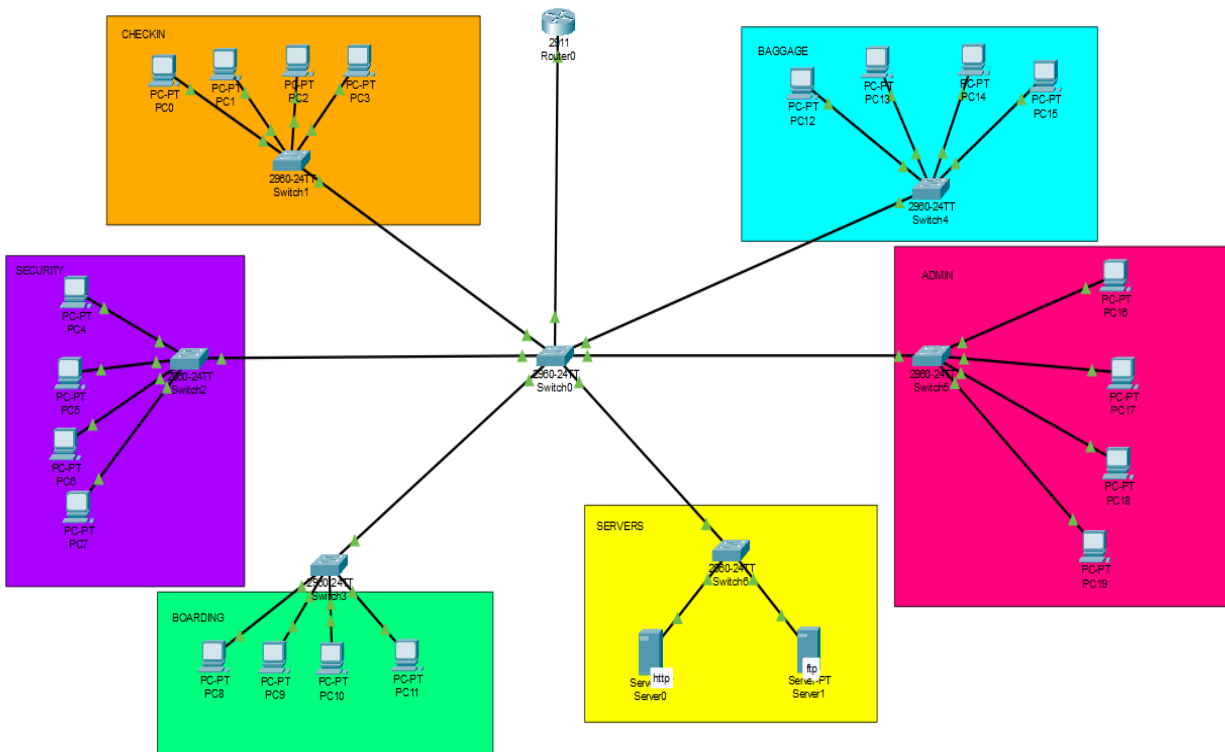


Fig 1.1 Airplane Network Design

6. Conclusion

The airplane network was implemented successfully using VLAN segmentation and trunk links. The network provides organized communication between departments and access to shared server services.